

Directions: Read all the questions carefully and answer them completely. No credit will be given for undocumented responses. **Show all your work to get full credit.** Give **exact answers** to questions unless explicitly requested otherwise.

1. Let

$$A = \begin{bmatrix} 1 & 5 \\ -5 & 2 \end{bmatrix}, B = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}.$$

Calculate AB^T and B^3 .

2. Use **Gaussian Elimination** or **Gauss-Jordan elimination method** to solve the linear system

$$\begin{aligned} 4x - 12y - 36z &= 0 \\ 3x + 9y - 27z &= 0 \end{aligned}$$

3. Find the inverse of AB if

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} \text{ and } B^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 0 \end{bmatrix}.$$

4. Find the inverse of the matrix

$$A = \begin{bmatrix} 7 & 2 & 2 \\ 10 & 3 & 3 \\ -6 & -2 & -1 \end{bmatrix}.$$

Verify the result.

5. Calculate the determinant of $A = \begin{bmatrix} 2 & 0 & 2 & 0 \\ 10 & 4 & -1 & 1 \\ -2 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$.

6. Write $B = \begin{bmatrix} 2 & 3 \\ -5 & 4 \end{bmatrix}$ as a product of two elementary matrices.

7. Find the L-U factorization of $A = \begin{bmatrix} 2 & 0 & 2 \\ 10 & 4 & -1 \\ -2 & 0 & 1 \end{bmatrix}$.